



NORTHERN UNIVERSITY

OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering



SMART FARMER ASSISTANT

AI-Based Agriculture Support System



Submitted To:

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Section: 8D

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CSE4204 — Mobile Computing Lab

Week 04 Submission

System Design &
Software Architecture

Smart Farmer Assistant

AI-Based Agriculture Support System

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1. Team Information

Field	Details
Team Name	CSE4204-8D-T03
Section	8D
Course	CSE4204 — Software Engineering Lab
Semester	Spring 2026
Institution	Leading University / University (Bangladesh)
GitHub Repo	github.com/sakib-foysal/CSE4204-8D-T03-SmartFarmer-Assistant

1.1 Team Members

#	Name	Student ID	Role
1	Sakib Foysal Ejarder	11220320948	Team Leader / Backend
2	Md. Noyon Sheikh	11220320946	Frontend Developer
3	Sayed Tauhidul Islam	11220320950	AI / ML Engineer
4	Sayed Akib Osman	11220320974	Database / QA

2. Project Information

Project Overview

Smart Farmer Assistant is an AI-powered platform designed to assist farmers across Bangladesh in making informed agricultural decisions. Agriculture is a critical sector for Bangladesh' economy, yet farmers face persistent challenges including crop disease losses (20-30% annual yield loss), improper fertilizer usage, limited access to qualified agricultural extension officers, and lack of real-time market information.

This system integrates multiple AI and IoT technologies into a single responsive web application accessible from any smartphone or web browser, providing instant crop disease detection, 24/7 AI chatbot support, real-time weather forecasting with flood alerts, live market price tracking, and intelligent fertilizer recommendations.

Project Scope

- Current Features:
 - AI Crop Disease Detection via image analysis
 - AI-powered Chatbot (Gemini AI) with Bangla support
 - Real-time Weather Forecasting and Flood Alerts
 - Live Market Price Tracking by region
 - Intelligent Fertilizer Recommendation Engine
 - User Registration, Login, and Profile Management
 - Disease Detection History and Chat History Tracking
 - Admin Panel for system management

Target Users

User Type	Description
Farmers (Primary)	Small and medium-sized farmers who need crop disease diagnosis and farming guidance
Agricultural Workers	Farm laborers who can benefit from disease detection and weather alerts
Admin Users	System administrators responsible for managing users and updating market data

3. UI/UX Design - Wireframes and Screens

The following section describes the major user interface screens of the Smart Farmer Assistant. The design follows a mobile-first responsive approach, ensuring optimal experience across all devices.

Design Philosophy

Simplicity: Clean, intuitive interface requiring minimal learning curve for rural farmers

Accessibility: Large touch targets, readable fonts, and bilingual (English/Bangla) support

Responsiveness: Works seamlessly on mobile phones, tablets, and desktop browsers

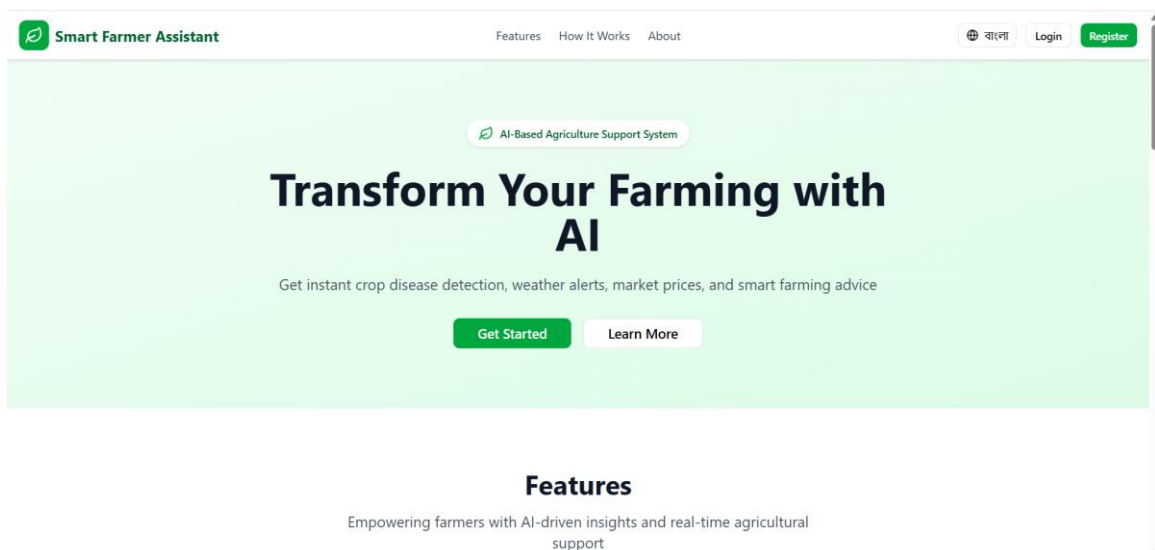
Performance: Fast loading times and minimal data consumption for low-bandwidth areas

Trust: Professional design that builds confidence in AI recommendations

3.1. Landing/Home Page

Welcome page showcasing key features of the system. Contains navigation menu, hero section with call-to-action buttons (Get Started, Learn More), and feature highlights.

- Key Elements:
 - Navigation Bar
 - Hero Section
 - Features Overview
 - Call-to-Action Buttons
 - Footer



How It Works

- 1 Register Your Account**
Sign up with your basic information to get started with the platform.
- 2 Upload Crop Images or Ask Questions**
Take a photo of your crop or ask our AI chatbot any farming question.
- 3 Get AI-Powered Recommendations**
Receive instant disease detection, treatment advice, fertilizer recommendations, and more.
- 4 Track Your Farming History**
Review past detections and conversations to make better farming decisions.

About

Smart Farmer Assistant is an AI-powered platform designed to help Bangladeshi farmers make better agricultural decisions. Agriculture accounts for a significant share of Bangladesh's GDP, yet farmers face persistent challenges — crop diseases cause 20–30% annual yield loss, improper fertilizer use degrades soil quality, and access to qualified agricultural extension officers remains limited in rural areas.

Our platform integrates AI image classification, a Gemini AI-powered chatbot, real-time weather forecasting, flood alerts, live market prices, and intelligent fertilizer recommendations into a single responsive platform accessible from any smartphone or web browser.

Ready to Transform Your Farming?

Join thousands of farmers already using AI to improve their crop yield and reduce losses.

Get Started



Smart Farmer Assistant

AI-Based Agriculture Support System



Features

Disease Detection
AI Chatbot
Weather & Flood Alert
Market Prices
Fertilizer Recommendation

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+880 1700-000000

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CSE4204-8D-T03 - Software Engineering Project



AI Disease Detection

Upload crop images and get instant disease diagnosis with treatment recommendations



Smart Chatbot

24/7 AI-powered farming assistant answering your questions in Bangla and English



Weather & Flood Alerts

Real-time weather forecasts and flood risk notifications for your location



Live Market Prices

Track real-time crop prices by region to sell at optimal times



Fertilizer Guide

Get AI-tailored fertilizer recommendations based on your crop and soil conditions



History Tracking

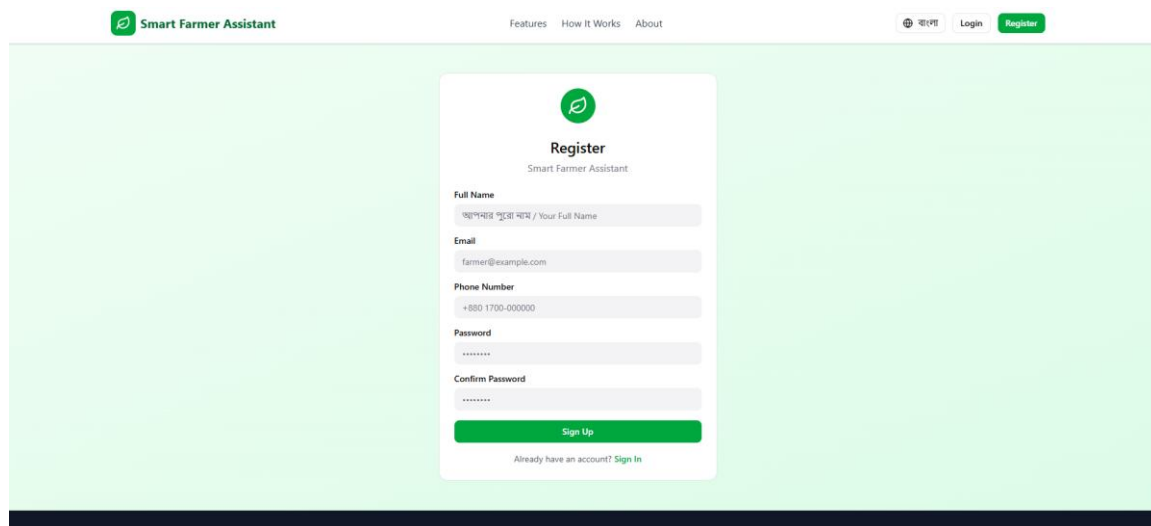
View your past disease detections and chat conversations

Press the "Print Screenshot" on your keyboard to take a screenshot

3.2. Registration Page

User registration form for new farmers. Collects full name, email, phone number, and password. Includes validation and error handling. Provides link to login page for existing users.

- Key Elements:
 - Full Name Field
 - Email Field
 - Phone Number Field
 - Password Fields (with confirmation)
 - Sign Up Button
 - Login Link

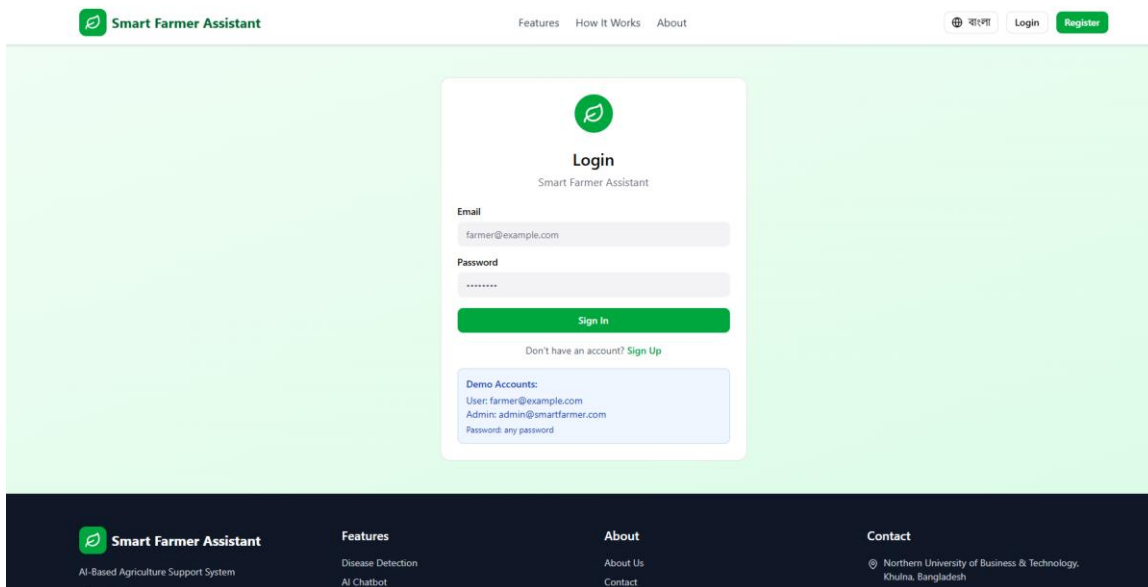


The screenshot displays the registration interface for the 'Smart Farmer Assistant' application. The page features a light green background. At the top, there is a navigation bar with the application logo on the left, and links for 'Features', 'How It Works', and 'About' in the center. On the right side of the navigation bar, there are links for 'Login' and 'Register'. The 'Register' link is highlighted with a green button. The main content area is a white card with a green header containing the application logo and the title 'Register'. Below the title, the text 'Smart Farmer Assistant' is displayed. The form includes several input fields: 'Full Name' (with a placeholder in Bengali and English), 'Email' (with a placeholder 'farmer@example.com'), 'Phone Number' (with a placeholder '+880 1700-000000'), 'Password' (with a placeholder '*****'), and 'Confirm Password' (with a placeholder '*****'). A green 'Sign Up' button is positioned below the password fields. At the bottom of the card, there is a link that says 'Already have an account? Sign In'.

3.3. Login Page

Authentication page with email and password fields. Includes demo account credentials for testing. Shows both Bangla and English text.

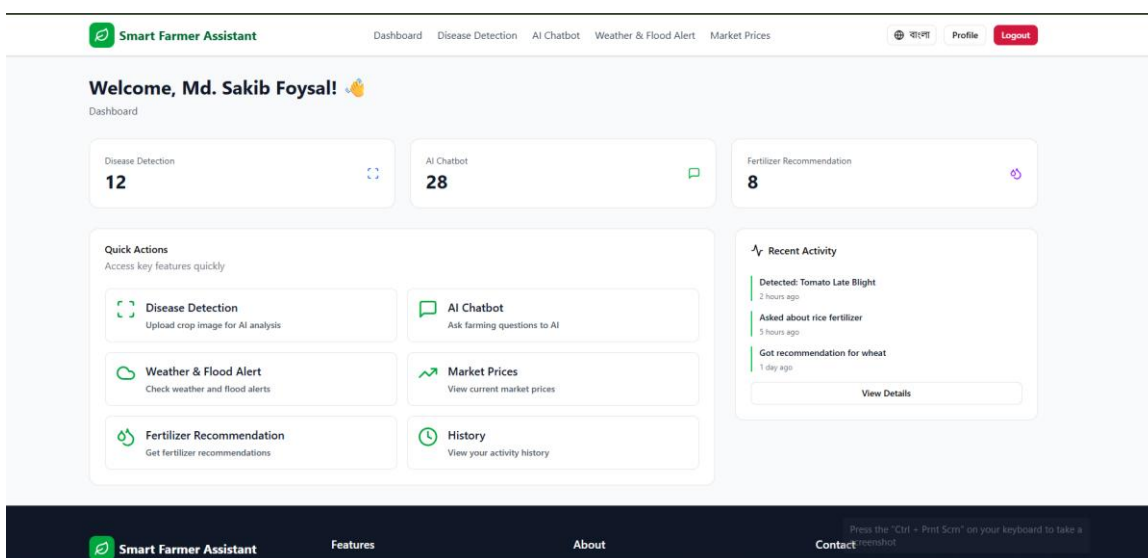
- Key Elements:
 - Email Input
 - Password Input
 - Sign In Button
 - Registration Link
 - Demo Credentials Section



3.4. Dashboard

Main landing page after login. Shows welcome message with user's name, quick statistics (disease detections, chatbot messages, fertilizer recommendations), quick action buttons, and recent activity feed.

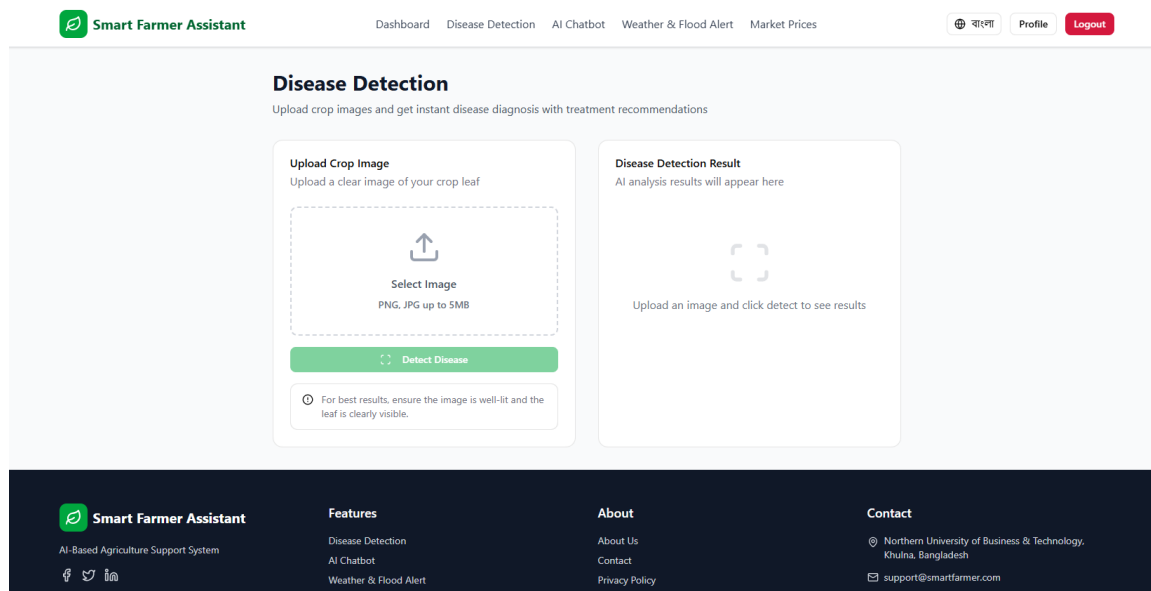
- Key Elements:
 - Welcome Message
 - Stats Cards
 - Quick Actions
 - Recent Activity
 - Navigation Menu



3.5. Disease Detection Page

Two-panel layout: left side for image upload with drag-and-drop support, right side for displaying AI analysis results. Shows disease name, confidence score, treatment recommendations.

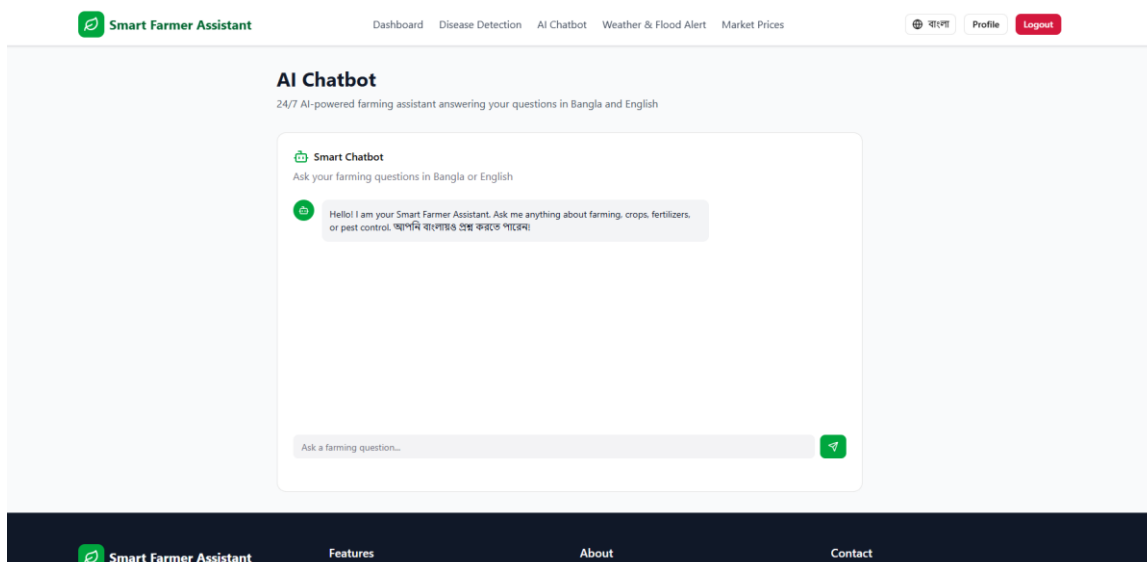
- Key Elements:
 - Image Upload Area
 - Detect Button
 - Results Panel
 - Disease Name
 - Confidence Score
 - Treatment Advice



3.6. AI Chatbot Interface

Chat-based interface for asking farming questions. Supports both Bangla and English. Shows conversation history. Displays AI responses with contextual farming advice.

- Key Elements:
 - Chat History
 - Question Input
 - Send Button
 - Language Toggle
 - AI Response Display

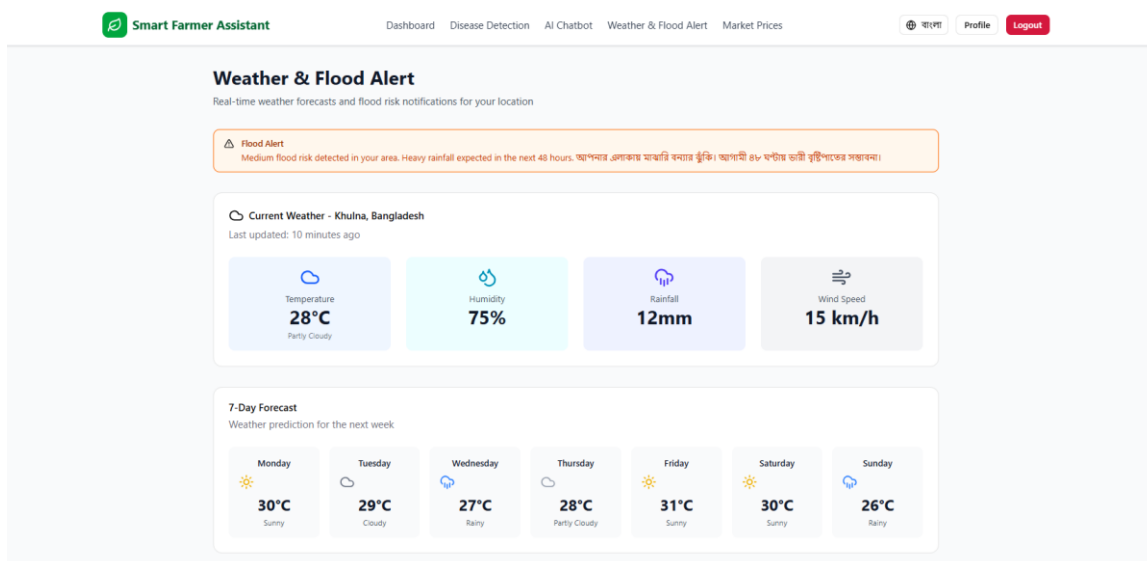


3.7. Weather & Flood Alert Page

Displays current weather with temperature, humidity, rainfall, wind speed. Shows 7-day forecast with daily weather icons. Includes flood risk alerts at the top.

- Key Elements:

- Flood Alert Box
- Current Weather Cards
- 7-Day Forecast
- Weather Icons
- Location Display

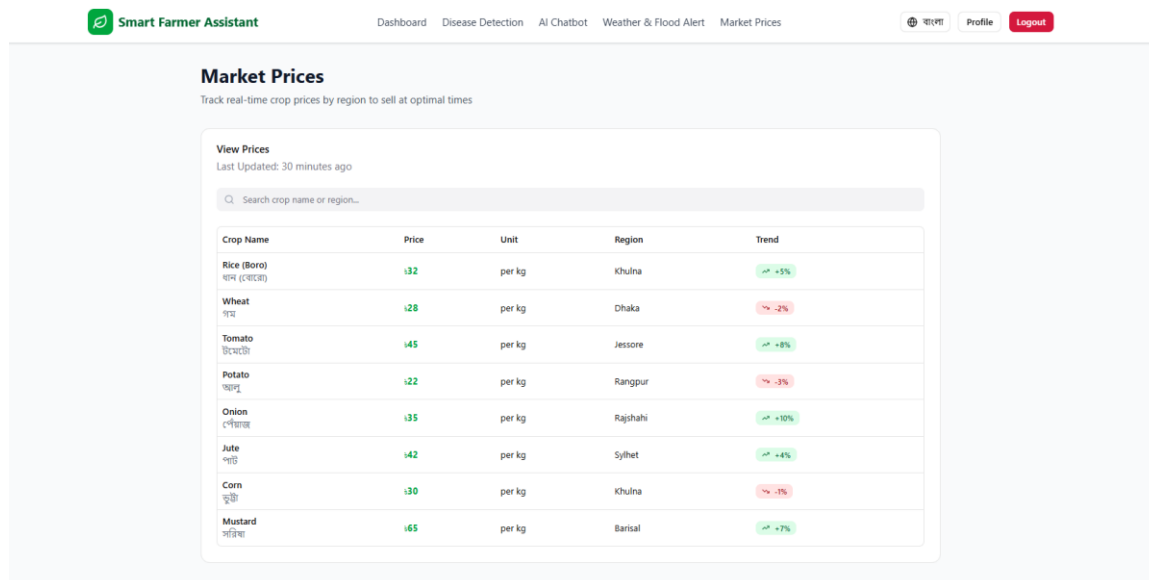


3.8. Market Prices Page

Table showing real-time crop prices by region. Includes price trends (up/down indicators).

Searchable and filterable by crop name or region.

- Key Elements:
 - Search Bar
 - Crop Name
 - Price
 - Unit
 - Region
 - Trend Indicator
 - Last Updated Time



Market Prices				
Track real-time crop prices by region to sell at optimal times				
View Prices Last Updated: 30 minutes ago				
Search crop name or region...				
Crop Name	Price	Unit	Region	Trend
Rice (Boro) ধান (বরো)	৳132	per kg	Khulna	↗ +5%
Wheat গম	৳128	per kg	Dhaka	↘ -2%
Tomato টম্যাটো	৳145	per kg	Jessore	↗ +8%
Potato আলু	৳122	per kg	Rangpur	↘ -3%
Onion পেঁপেঁড়া	৳135	per kg	Rajshahi	↗ +10%
Jute পাট	৳142	per kg	Sylhet	↗ +4%
Corn চুড়ী	৳130	per kg	Khulna	↘ -1%
Mustard সরিষা	৳165	per kg	Barisal	↗ +7%

3.9. User Profile Page

User account management page. Shows profile avatar with initials, personal information, account statistics (member since, total detections, chat messages), edit profile button, and change password section.

- Key Elements:
 - Profile Avatar
 - User Name
 - Email
 - Phone
 - Account Stats

- Edit Profile Button
- Change Password Section

Smart Farmer Assistant Dashboard Disease Detection AI Chatbot Weather & Flood Alert Market Prices [বাংলা](#) [Profile](#) [Logout](#)

Profile

Manage your account information

Md. Sakib Foysal Ejarder
sakib@example.com
Farmer Account

Member since	Jan 2024
Total Detections	12
Chat Messages	28

Profile Information
Update your personal details

Full Name
Md. Sakib Foysal Ejarder

Email
sakib@example.com

Phone Number
+880 1700-123456

[Edit Profile](#)

Change Password
Update your password to keep your account secure

Current Password
[password field]

New Password
[password field]

Confirm Password
[password field]

[Update Password](#)

3.10. Admin Page

Main control panel for administrators after login. Displays a welcome message with the admin's name, system overview statistics (total users, disease reports, fertilizer recommendations, chatbot interactions), quick management actions, notifications, and recent system activities for efficient monitoring and administration.

Smart Farmer Assistant

DashboardUser ManagementUpdate Market Prices

বাংলাProfileLogout

Admin Panel

System Activity Overview

Total Users

1,234

+12%

Total Detections

5,678

+8%

Total Chats

12,345

+15%

Active Today

342

+5%

Weekly Activity

User engagement metrics for the past week

Mon

Tue

Wed

Thu

Fri

Sat

Sun

Recent Users

Latest registered farmers

Md. Karim

karim@example.com

2 hours ago

active

Fatema Begum

fatema@example.com

5 hours ago

active

Abdul Rahman

abdul@example.com

1 day ago

active

Hasina Khatun

hasina@example.com

2 days ago

inactive

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AI Chatbot

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User Management

Manage registered farmers and their accounts

Total Users: 6

All registered farmer accounts

Search by name, email or phone...

ID	Name	Email	Phone	Joined	Status	Actions
#1	Md. Karim Ahmed	karim@example.com	+880 1700-111111	2024-01-15	Active	
#2	Fatema Begum	fatema@example.com	+880 1700-222222	2024-02-20	Active	
#3	Abdul Rahman	abdul@example.com	+880 1700-333333	2024-03-10	Active	
#4	Hasina Khatun	hasina@example.com	+880 1700-444444	2024-04-05	Inactive	
#5	Rahim Miah	rahim@example.com	+880 1700-555555	2024-05-12	Active	
#6	Ayesha Siddika	ayesha@example.com	+880 1700-666666	2024-06-01	Active	

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Smart Farmer Assistant

DashboardUser ManagementUpdate Market Prices

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Update Market Prices

Update market prices for farmers

Current Market Prices

Manage all crop prices across regions

+ Add Price

ID	Crop Name	Price	Unit	Region	Actions
#1	Rice (Boro)	৳32	per kg	Khulna	
#2	Wheat	৳28	per kg	Dhaka	
#3	Tomato	৳45	per kg	Jessore	
#4	Potato	৳22	per kg	Rangpur	
#5	Onion	৳35	per kg	Rajshahi	

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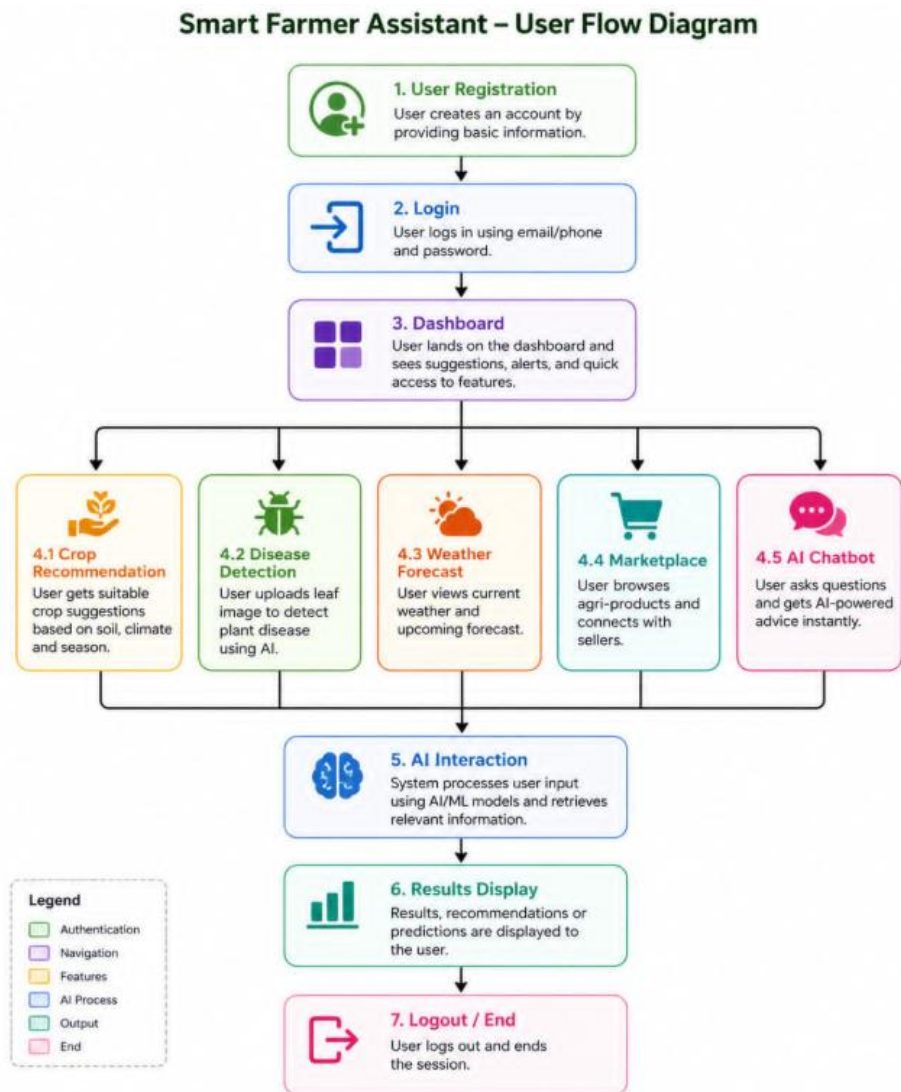
+880 1700 000000

Design Features

- Bilingual Support (Bangla & English) for accessibility
- Green color scheme (#1DA758) for agricultural/environmental theme
- Responsive grid layout optimized for mobile (320px) to desktop (1920px)
- Consistent navigation bar with language toggle and user menu
- Large, touch-friendly buttons for rural users
- Clear form validation and error messages
- Loading indicators for AI processing
- Confirmation dialogs for destructive actions

4. User Flow Design

The following diagrams and descriptions illustrate how users navigate through the Smart Farmer Assistant system.



Main User Journey Flow

Step	Description
1. Application Start	User opens the application/website
2. Authentication	User navigates to Login or Registration page

3. Credentials Entry	User enters email and password (or registration details)
4. System Verification	Backend validates credentials and creates/verifies JWT token
5. Dashboard Access	Upon successful authentication, user enters the Dashboard
6. Service Selection	User selects desired service from quick actions or menu
7. Feature Interaction	User interacts with selected feature (Disease Detection, Chatbot, etc.)
8. AI Processing	System processes request and queries AI services
9. Results Display	Results are displayed to the user with recommendations
10. History Save	User interactions are saved to database
11. End or Continue	User can perform another action or logout

Feature-Specific Flows

- A. Disease Detection Flow:
 - User opens Disease Detection page
 - User uploads a clear crop leaf image (PNG/JPG, max 5MB)
 - System validates image format and size
 - Image is pre-processed (resized to 224×224 pixels)
 - TensorFlow CNN model performs inference
 - If confidence $\geq 75\%$, disease name is returned; otherwise "Unidentified"
 - Gemini AI generates treatment recommendations
 - Fertilizer recommendation is also generated
 - Complete report is displayed with disease, confidence, treatment, and fertilizer
 - User can save, print, or share the report
 - Record is saved to database for history
- B. Chatbot Interaction Flow:

- User opens AI Chatbot page
 - System displays greeting message and instructions
 - User types a farming question in Bangla or English
 - User selects language (if not auto-detected)
 - Backend adds agriculture context to the question
 - Gemini AI API is called with the enriched prompt
 - AI generates contextual, farming-specific response
 - Response is displayed in the chat UI
 - Question and response are saved to chat_history
 - User can ask follow-up questions or end conversation
 - Conversation history is preserved for future reference
-
- C. Weather & Flood Alert Flow:
 - User opens Weather & Flood Alert page
 - System automatically detects user's location
 - Backend calls OpenWeatherMap API for current weather
 - Backend checks flood risk data from weather API
 - Current weather metrics are displayed (temp, humidity, rainfall, wind)
 - 7-day forecast is retrieved and displayed with weather icons
 - If flood risk is detected, alert is prominently shown at top
 - User can receive push notifications for severe weather
 - Data is cached and updated every 10 minutes

5. Development Roadmap

The development roadmap outlines the sequence of work to be completed over the remaining weeks of the semester. The project follows an iterative development approach with weekly milestones.

Phase-Based Roadmap

Phase 1: Backend API Development

Timeline: Week 6

Lead: Sayed Tauhidul Islam

- Tasks:
 - Implement all REST API endpoints (authentication, disease detection, chatbot, weather, market, fertilizer)
 - Set up JWT authentication and authorization middleware
 - Configure Django settings for development and production
 - Implement image upload handling and validation
 - Create database migrations and initialize PostgreSQL
 - Set up error handling and logging
 - Implement rate limiting for API endpoints
 - Test all endpoints with Postman

Phase 2: Frontend Development

Timeline: Week 7

Lead: Sayed Akib Osman

- Tasks:
 - Complete all React components for all pages
 - Implement responsive design using Tailwind CSS
 - Set up React Router for navigation
 - Implement state management (Context API or Redux)
 - Create API integration layer for backend calls
 - Implement authentication flow with JWT tokens
 - Add bilingual support (Bangla/English) with i18n library
 - Implement form validation on frontend

- Add loading states and error handling
- Optimize images and assets

Phase 3: AI Model Integration

Timeline: Week 8

Lead: Md. Noyon Sheikh

- Tasks:
 - Fine-tune TensorFlow CNN model on PlantVillage dataset
 - Create image preprocessing pipeline in backend
 - Integrate TensorFlow model into Django backend
 - Set up Gemini AI API integration for chatbot
 - Implement prompt engineering for agricultural context
 - Create fertilizer recommendation logic using rule engine
 - Test AI predictions with sample images
 - Optimize model for inference speed (<5 seconds)
 - Implement confidence threshold logic (75%)
 - Create fallback mechanisms for unidentified diseases

Phase 4: External API Integration

Timeline: Week 8

Lead: Md. Noyon Sheikh

- Tasks:
 - Integrate OpenWeatherMap API for weather data
 - Set up weather data caching and update schedule
 - Implement flood risk detection logic
 - Integrate market price API for crop pricing
 - Set up scheduled tasks for data refresh
 - Implement location-based services
 - Test API integrations with real data
 - Implement error handling for API failures
 - Set up monitoring for API health

Phase 5: Feature Completion

Timeline: Week 9

Lead: Sakib Foysal Ejarder

- **Tasks:**
 - Feature Completion Complete implementation of all core system features.
 - Final integration of Frontend, Backend, Database, and AI modules.
 - Complete User Authentication (Registration, Login, Logout, Profile Management).
 - Finalize AI Crop Disease Detection with image upload and prediction.
 - Complete AI Chatbot with Bangla and English language support.
 - Complete Weather Forecast and Flood Alert module.
 - Integrate Live Market Price Tracking. Finalize Fertilizer Recommendation System.
 - Complete User Dashboard and History Management. Implement Admin Panel functionalities. Fix remaining bugs and improve application performance.
 - Conduct final integration testing and user acceptance testing (UAT).
 - Optimize UI/UX for mobile and desktop responsiveness.
 - Prepare the system for deployment and final demonstration.
 - Update project documentation and GitHub repository.

Phase 6: Testing & Debugging

Timeline: Week 10

Lead: Sakib Foysal Ejarder

- **Tasks:**
 - Write unit tests for backend endpoints
 - Write integration tests for API flows
 - Perform end-to-end testing of all features
 - Test responsive design on multiple devices
 - Perform security testing (SQL injection, CSRF, etc.)
 - Test AI model accuracy and performance
 - User acceptance testing with sample farmers
 - Bug fixing and optimization
 - Performance testing and optimization

- Document known issues and workarounds

Phase 7: Deployment & Documentation

Timeline: Week 11

Lead: Sakib Foysal Ejarder

- Tasks:
 - Deploy frontend to Vercel
 - Deploy backend to Render
 - Set up PostgreSQL database on Neon
 - Configure environment variables for production
 - Set up SSL certificates and HTTPS
 - Implement monitoring and logging
 - Create API documentation with Swagger/OpenAPI
 - Write user documentation and guides
 - Prepare deployment checklist
 - Perform smoke testing on production

Phase 8: Final Polish & Presentation

Timeline: Week 12

Lead: Sakib Foysal Ejarder | Sayed Tauhidul Islam | Md Noyon Sheikh | Sayed Akib Osman

- Tasks:
 - Create project demo video
 - Prepare presentation slides
 - Write final project report
 - Update GitHub documentation
 - Create video tutorial for users
 - Prepare demo accounts for presentation
 - Test demo flow thoroughly
 - Create handover documentation

6. Team Task Distribution

Each team member has been assigned specific responsibilities and modules to develop. Clear ownership ensures accountability and efficient parallel development.

Team Member Responsibilities

Sayed Tauhidul Islam

Student ID: 11220320950

Role: Backend Developer

- Responsibilities:
 - Overall project management and coordination
 - Backend API architecture and design
 - Authentication and authorization system (JWT)
 - REST API endpoint implementation
 - Database design and migrations
 - Backend testing and debugging
 - Deployment and DevOps
 - Documentation and reporting
 - Integration of all backend modules

Sayed Akib Osman

Student ID: 11220320974

Role: Frontend Developer

- Responsibilities:
 - Frontend architecture and component design
 - React.js application development
 - UI implementation based on wireframes
 - Responsive design using Tailwind CSS
 - Navigation and routing implementation
 - State management implementation

- API integration on frontend
- Form handling and validation
- User authentication flow
- Bilingual interface implementation

Md. Noyon Sheikh

Student ID: 11220320948

Role: AI/ML Engineer

- Responsibilities:
 - TensorFlow model selection and training
 - Image preprocessing pipeline development
 - AI model integration in backend
 - Gemini AI chatbot integration
 - Prompt engineering for agricultural context
 - Fertilizer recommendation logic
 - Model performance optimization
 - Accuracy testing and validation
 - Handling edge cases and errors
 - AI model documentation

Sakib Foysal Ejarder

Student ID: 11220320974

Role: Team Leader / Database Engineer

- Responsibilities:
 - Database design and optimization
 - PostgreSQL setup and configuration
 - External API integrations (OpenWeatherMap, Market Prices)
 - Data validation and integrity
 - Quality assurance and testing
 - Performance testing and optimization
 - Security testing

- Bug documentation and tracking
- Test case creation
- Monitoring and maintenance

Module Ownership

Module	Owner(s)
User Authentication & Authorization	Sakib Foysal Ejarder
Disease Detection API & AI	Md Noyon Sheikh, Sayed Tauhidul Islam, Sakib Foysal Ejarder
AI Chatbot Integration	Md Noyon Sheikh, Sakib Foysal Ejarder
Weather & Flood Alert Module	Sayed Tauhidul Islam, Sakib Foysal Ejarder
Market Prices Module	Sayed Tauhidul Islam, Sakib Foysal Ejarder
Fertilizer Recommendation	Md Noyon Sheikh, Sakib Foysal Ejarder
Frontend UI/UX Implementation	Sayed Akib Osman
Database Design & Management	Sayed Akib Osman, Sakib Foysal Ejarder
Testing & QA	Sakib Foysal Ejarder

7. Final Technology Stack

The Smart Farmer Assistant uses a modern, scalable technology stack optimized for agricultural applications with strong AI/ML capabilities.

Frontend

React.js: v18.x - UI library for interactive components

Tailwind CSS: v3.x - Utility-first CSS framework for responsive design

JavaScript/ES6+: Modern JavaScript for client-side logic

React Router: Navigation and routing

Axios: HTTP client for API calls

i18n: Internationalization for Bangla/English support

Context API: State management solution

Vercel: Deployment platform for frontend

Backend

Django: v4.x - Python web framework

Django REST Framework: RESTful API development

Python: v3.9+ - Server-side programming language

JWT (PyJWT): Token-based authentication

Celery: Asynchronous task queue

Redis: Caching and task broker

Render: Backend deployment platform

Database

PostgreSQL: v14+ - Relational database

Neon: Serverless PostgreSQL hosting

SQLAlchemy: ORM for database operations

Alembic: Database migration tool

AI & Machine Learning

TensorFlow: v2.x - Deep learning framework for disease detection

Keras: High-level API for neural networks

OpenCV: Image processing library

NumPy: Numerical computing

Pandas: Data manipulation and analysis

Scikit-learn: Machine learning utilities

Google Gemini API: Chatbot and recommendation AI

PlantVillage Dataset: Training data for disease classification

External APIs

OpenWeatherMap API: Real-time weather and flood data

Custom Market Price API: Crop price data integration

Google Gemini API: AI-powered responses

Development Tools

Git/GitHub: Version control

VS Code: Code editor

Postman: API testing

Docker: Containerization (optional for deployment)

npm/pip: Package managers

GitHub Actions: CI/CD pipeline

Testing

pytest: Python testing framework

Django TestCase: Unit testing

Jest: JavaScript testing

React Testing Library: Component testing

Selenium: End-to-end testing (optional)

Infrastructure & DevOps

Vercel: Frontend hosting and CDN

Render: Backend hosting

Neon PostgreSQL: Database hosting

GitHub: Repository and CI/CD

SSL/HTTPS: Security

Environment Variables: Configuration management

Technology Justification

React.js: Popular, component-based UI library with large ecosystem and community support. Excellent for building responsive SPAs.

Django: Mature, batteries-included web framework with excellent ORM, authentication, and admin panel. Perfect for rapid API development.

PostgreSQL: Robust relational database with excellent support for complex queries and data integrity. Ideal for agricultural data.

TensorFlow: Industry-leading deep learning framework with pre-trained models available. Best choice for image classification tasks.

Gemini AI: Latest large language model with agricultural context awareness. Supports Bangla language well.

Tailwind CSS: Modern utility-first CSS framework enabling rapid, responsive UI development without custom CSS files.

Vercel/Render: Serverless platforms with automatic scaling, excellent for handling variable traffic from farmers.

8. Design Guidelines and Best Practices

UI/UX Principles

User-Centered Design: All design decisions prioritize farmer needs and accessibility.

Consistency: Consistent color scheme, typography, and component behavior throughout the application.

Simplicity: Minimal cognitive load with clear hierarchies and straightforward workflows.

Accessibility: WCAG 2.1 AA compliance with large touch targets, clear labels, and bilingual support.

Performance: Fast loading times and smooth interactions ensure good user experience on all devices.

Feedback: Clear visual feedback for all user actions (loading states, success messages, error messages).

Responsive Design: Mobile-first approach ensuring optimal experience on all screen sizes.

Color Scheme

Color	Hex Code	Usage
Primary Green	#1DA758	Buttons, headers, primary actions
Light Green	#D5F4E6	Secondary backgrounds, hover states
Dark Gray	#333333	Text content, headlines
Light Gray	#F5F5F5	Page backgrounds, card backgrounds
Red/Orange	#E74C3C	Alerts, errors, destructive actions

Typography

Font Family: Calibri / Arial (system fonts for web compatibility)

Element	Size	Weight	Usage
Page Title/H1	32px	Bold	Main page headings
Section Heading/H2	24px	Bold	Section titles
Subsection/H3	18px	Bold	Subsection titles

Body Text	14px	Regular	Main content text
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Responsive Breakpoints

Device	Width Range	Layout
Mobile	320px - 480px	Single column, full-width elements
Tablet	481px - 1024px	Two-column layout
Desktop	1025px+	Multi-column layout, optimized spacing

Design Do's and Don'ts

- DO:
 - Use consistent spacing (8px grid system)
 - Maintain clear visual hierarchy
 - Provide clear error messages with solutions
 - Use icons to improve scannability
 - Test on actual devices
 - Implement dark mode considerations
 - Optimize images for fast loading
 - Use semantic HTML
- DON'T:
 - Avoid excessive animations
 - Don't use colors as only indicator
 - Avoid cluttered layouts
 - Don't make touch targets too small (<44px)
 - Avoid auto-playing videos
 - Don't use outdated design patterns
 - Avoid unclear error messages
 - Don't forget accessibility

Conclusion

The Smart Farmer Assistant Week 5 documentation provides a comprehensive blueprint for the remaining development phases. With clear UI/UX designs, well-defined user flows, a structured development roadmap, and explicit task distribution among team members, we are positioned to deliver a high-quality, farmer-friendly agricultural support system.

The project leverages cutting-edge technologies (React, Django, TensorFlow, Gemini AI) while maintaining focus on accessibility, performance, and usability for rural farmers in Bangladesh. Our iterative approach, phased development, and comprehensive testing strategy will ensure that all features are properly implemented and thoroughly tested before final deployment.

By Week 9, the Smart Farmer Assistant will be ready for production deployment, helping thousands of farmers make better agricultural decisions through AI-powered insights and real-time information.